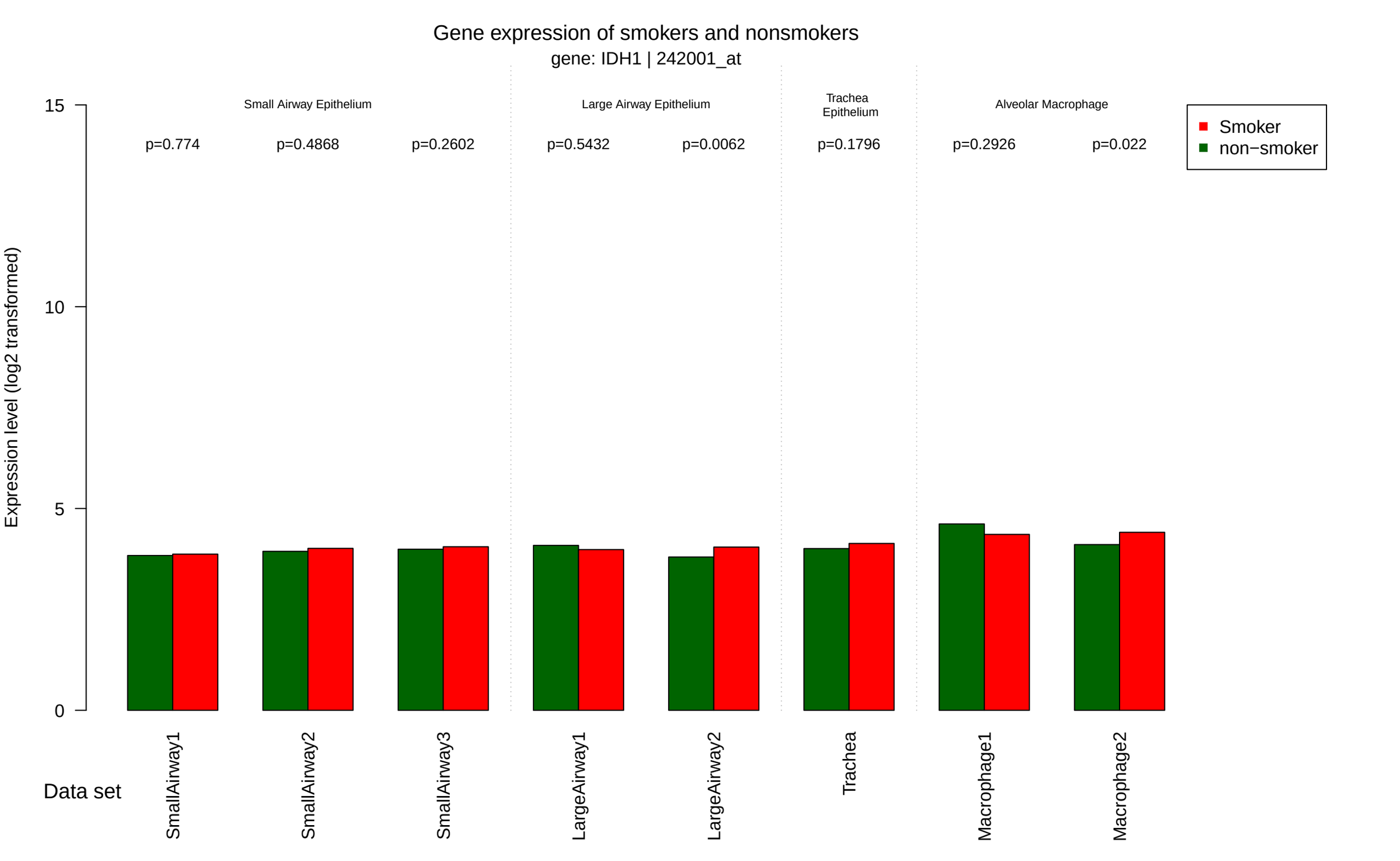
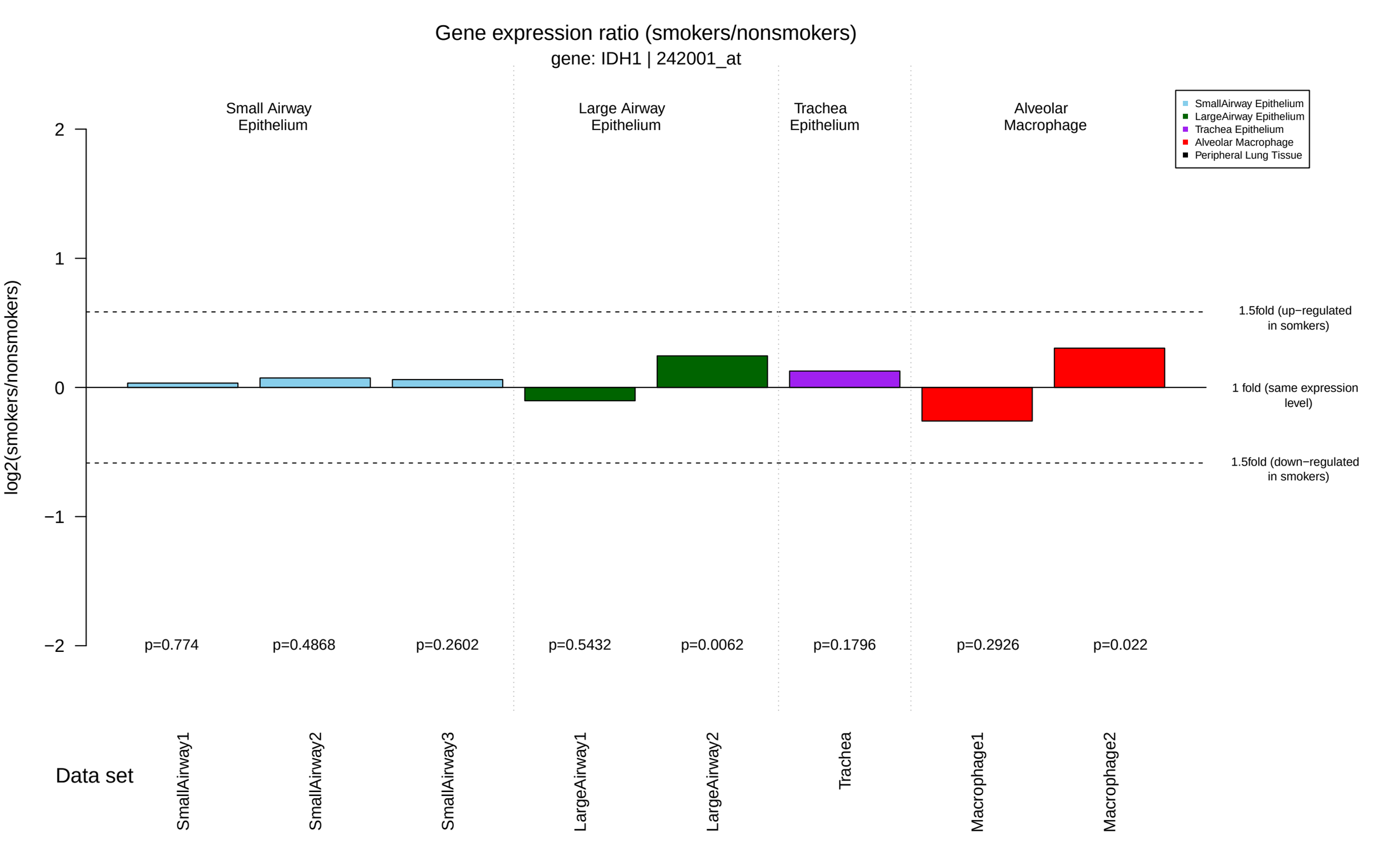
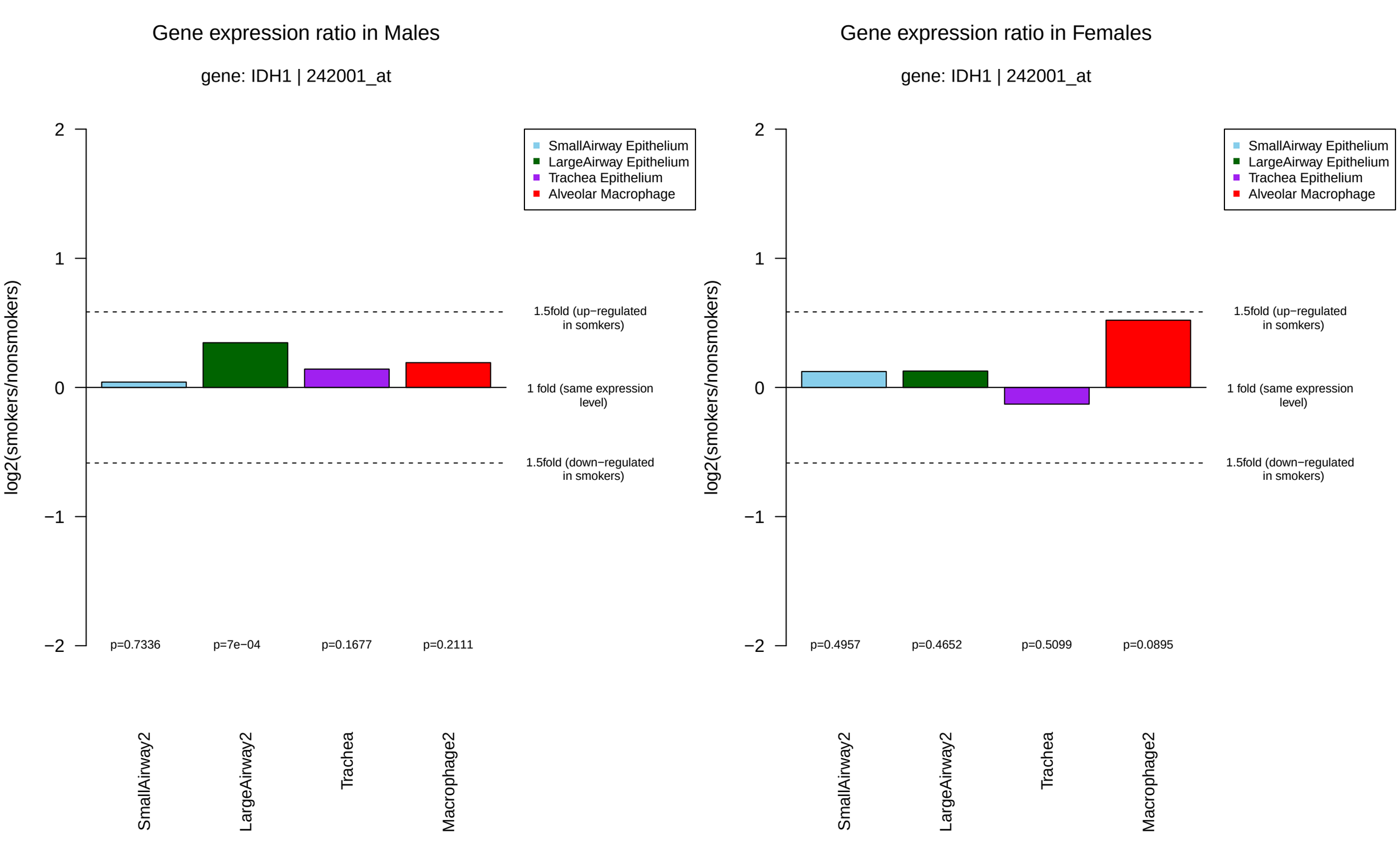
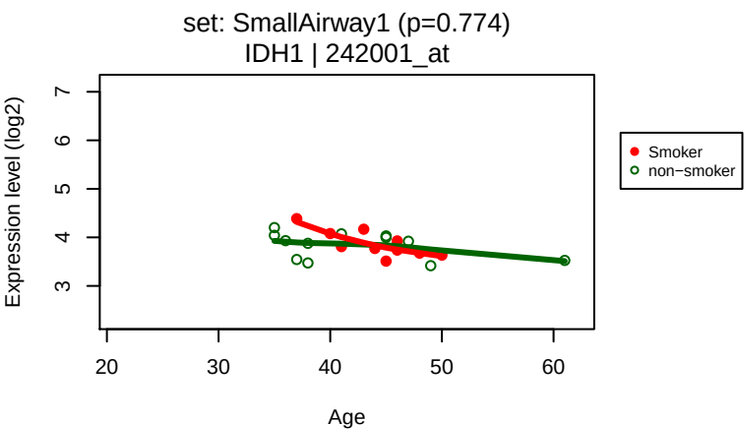
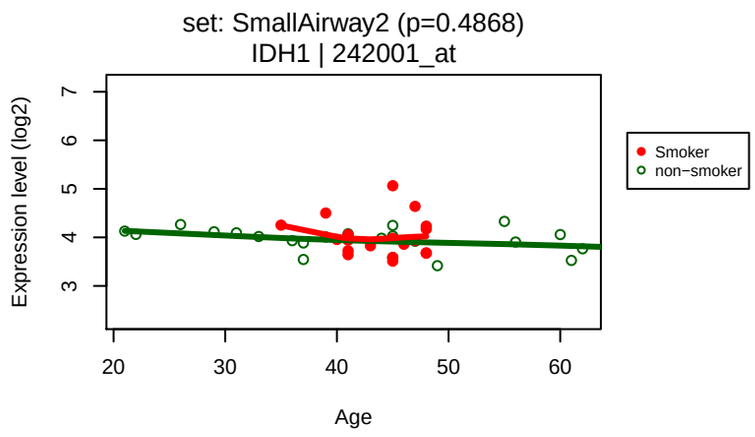
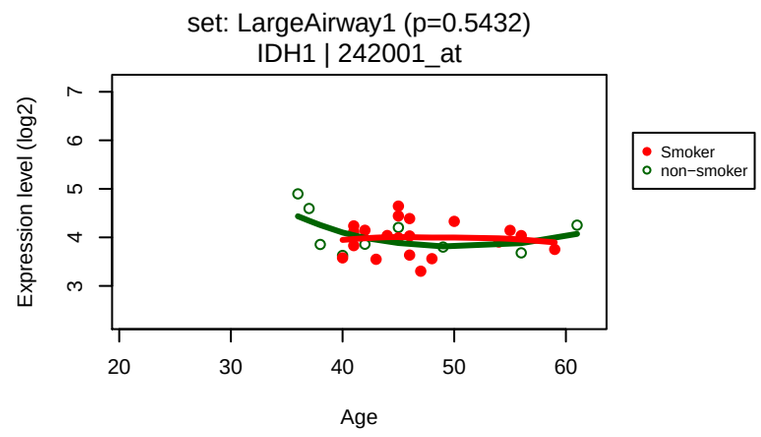

Gene expression of smokers and nonsmokers
gene: IDH1 | 242001_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.774
p=0.4868
p=0.2602
p=0.5432
p=0.0062
p=0.1796
p=0.2926
p=0.022
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

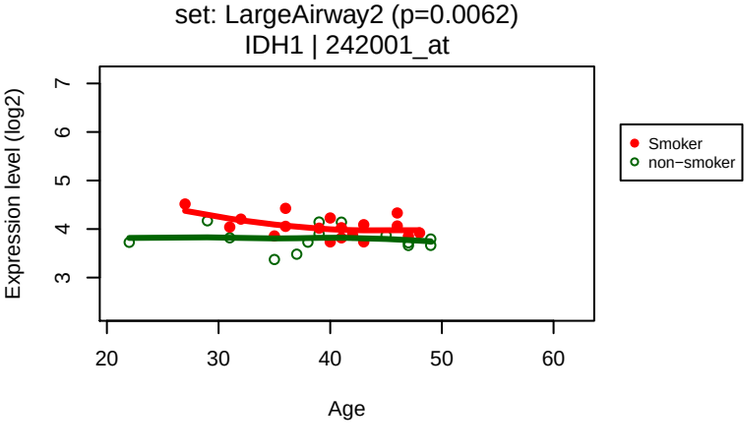

Gene expression ratio (smokers/nonsmokers)
gene: IDH1 | 242001_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.774
p=0.4868
p=0.2602
p=0.5432
p=0.0062
p=0.1796
p=0.2926
p=0.022
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

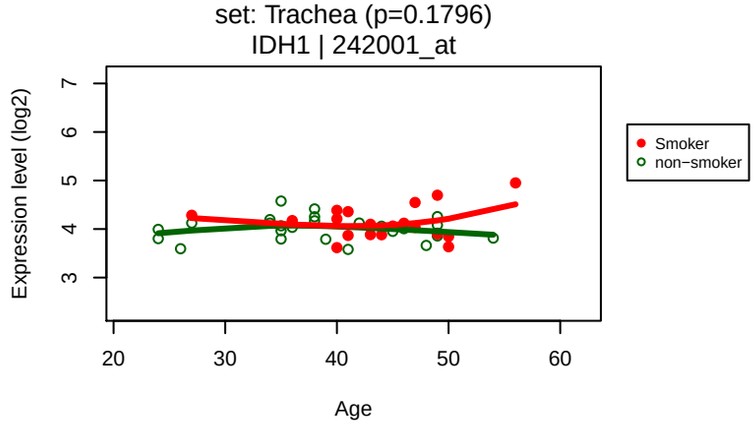

Gene expression ratio in Males
gene: IDH1 | 242001_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7336
p=7e−04
p=0.1677
p=0.2111
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IDH1 | 242001_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4957
p=0.4652
p=0.5099
p=0.0895
SmallAirway2
LargeAirway2
Trachea
Macrophage2

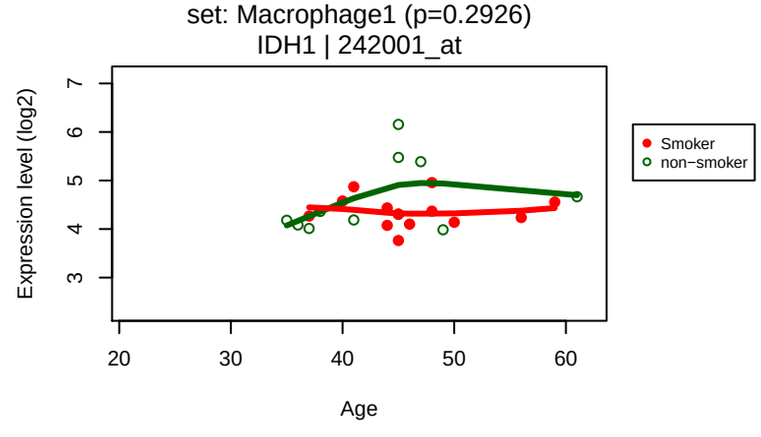

set: SmallAirway1 (p=0.774)
IDH1 | 242001_at
Expression level (log2)
Age
Smoker
non−smoker

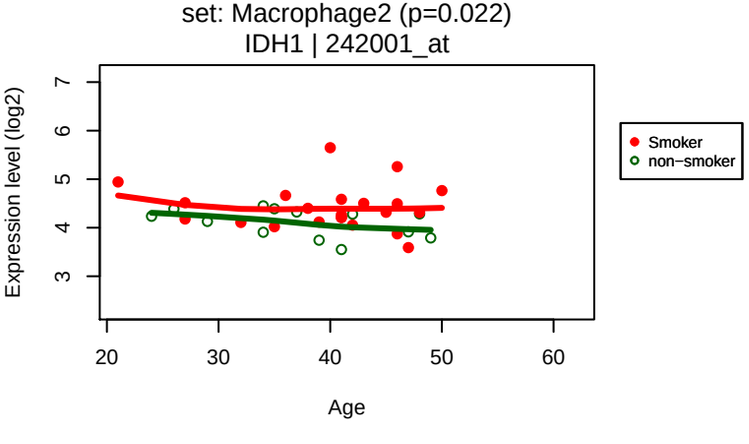

set: SmallAirway2 (p=0.4868)
IDH1 | 242001_at
Expression level (log2)
Age
Smoker
non−smoker

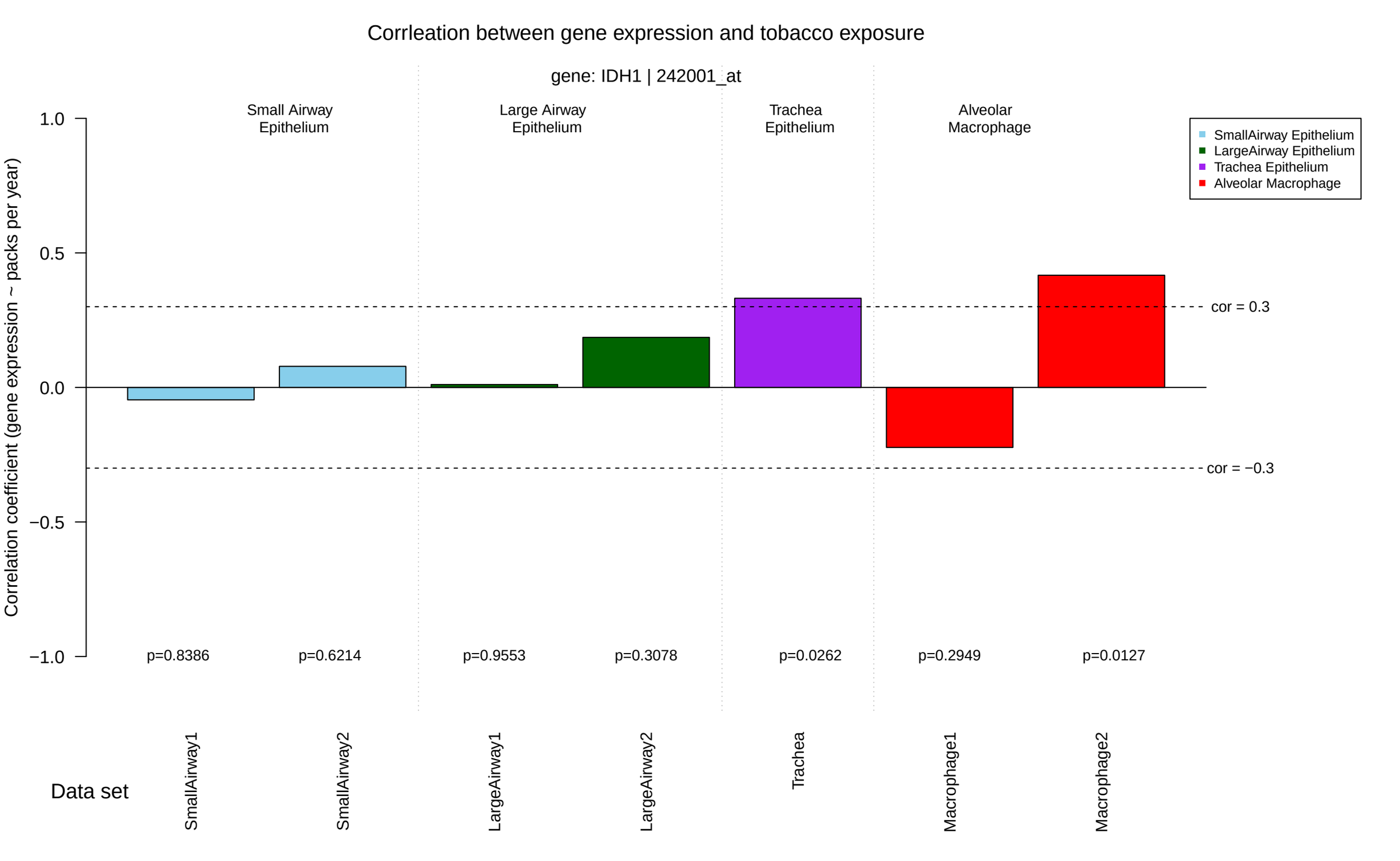

set: LargeAirway1 (p=0.5432)
IDH1 | 242001_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0062)
IDH1 | 242001_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1796)
IDH1 | 242001_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2926)
IDH1 | 242001_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.022)
IDH1 | 242001_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IDH1 | 242001_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8386
p=0.6214
p=0.9553
p=0.3078
p=0.0262
p=0.2949
p=0.0127
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2